OpenDC-LCA: from heterogeneous evidence to bounded cooling decisions

One calculation spine keeps the functional unit, boundary, transformation, uncertainty and claim status visible.

DECISION QUESTION

Which cooling architecture is preferred, where can the ranking change, and what evidence would resolve it?

1 Evidence with declared role

Released lifecycle model
Microsoft/WSP component results
Role: released-total reconciliation

Public scenario evidence
eGRID · Boavizta · ÖKOBAUDAT
Role: declared screening scenarios

Exchange-ready inventories
USLCI · GLAD · openLCA/Brightway
Role: exchange-tested; not in comparison

Primary engineering evidence
Performance maps · BOM · failures
Role: required for decision-grade claims

2 Harmonization and claim gate

1. **Define equivalence**
   Useful computation · lifetime · reliability
2. **Assess method alignment**
   Unit · GWP basis · year · geography
3. **Expose transformation limits**
   Interpolation · extrapolation · boundary gap
4. **Propagate and classify evidence**
   Joint stress · provenance · review status

SCREENING ≠ COMPARATIVE ASSERTION

3 Decision outputs

Reproducibility

24 released totals reconciled
source IDs + equations + digests

Released-model crossover

96.7 kg CO_2e/MWh
cold plate = single-phase

Screening limits and evidence needs

29.8% beyond released anchors

99% → 55%
two-phase first rank under narrow → wide stress
declared draws; not a probability

reconstructed · screening transformation · registered / unlinked · primary / decision-grade

Every numerical claim retains its evidence class.